

INTERIM REPORT

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**CHARACTERISTICS OF HUMAN INFLUENZA AH1N1, AH3N2 AND B
VIRUSES ISOLATED SEPTEMBER 2006 TO FEBRUARY 2007**

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During this period, influenza activity in the northern hemisphere has in general been moderate. Influenza AH3N2 viruses have predominated in most European countries and in some Asian countries. In contrast, influenza AH1N1 viruses have been more prevalent in the USA. Influenza B viruses have circulated at low levels.

Some 320 influenza viruses, isolated in 23 countries since September 2006, were characterized by the Centre. The majority were influenza A H3N2 (74%); H1N1 viruses accounted for 23% (Table 1). Few B viruses were received, including examples of both B/Victoria and B/Yamagata lineages.

AH1N1 viruses

H1N1 viruses were antigenically closely related to either the current vaccine strain A/New Caledonia/20/99 and more recent reference viruses, such as A/Thessaloniki/24/2005 and A/Egypt/39/05, or the recently recommended vaccine strain A/Solomon Islands/3/2006 and reference viruses, A/Hong Kong/2652/2006 and A/Fukushima/141/2006 (Table 2). Approximately equal numbers of the two antigenic variants were received, although the relative proportions varied between countries. For example, H1N1 viruses isolated in Madagascar, Morocco, The Netherlands and Russia were mainly A/New Caledonia/20/99-like, whereas viruses isolated in Italy and Norway were mainly A/Solomon Islands/3/2006-like.

As previously, the HA sequences fell into two principal clades (Figure 1). Clade 1, represented by A/Thessaloniki/24/2005, was characterized by the amino acid change Y252F, relative to the HA sequence of A/New Caledonia/20/99. There was relatively little change in these HA sequences or the corresponding NA sequences of viruses isolated during the past two years (Figures 1 and 2, Table 3). HA sequences of most recent isolates fell within Clade 2, which is characterized by the common amino acid changes T82K, Y94H, R145K, R208K and T266N. Clade 2 exhibits greater diversity and includes 3 principal sublineages, two of which are represented by the antigenic variant reference viruses, A/Hong Kong/2652/2006 and A/Solomon Islands/3/2006 (and A/Fukushima/141/2006) (Figure 1, Table 3). Despite extensive differences in HA sequences, there was a close correlation between the difference in antigenicity of A/Solomon Islands/3/2006-like (including A/HK/2652/2006-like and A/Fukushima/141/2006-like) viruses from A/New Caledonia/20/99-like (including A/Egypt/39/2006-like) viruses and the presence of the amino acid change K140E. Equivalent heterogeneity was observed in the divergence of corresponding sublineages of NA sequences (Figure 2). Common differences which characterized Clade 2 NAs included V234M and D382N, relative to the NA sequence of A/New Caledonia/20/99 (Figure 2, Table 3).

M2 protein gene sequences obtained for the majority of selected viruses with HAs in clades 1 and 2 did not contain amantadine resistance mutations. As previously, viruses with HAs which fell into the subgroup represented by A/HK/2652/06 possessed the resistance mutation S31N.

AH3N2 viruses

Many of the H3N2 viruses were antigenically closely related to the A/Wisconsin/67/2005-like MDCK cell-grown reference virus A/Hong Kong/4443/05,

although in general, HI titres were low, relative to the homologous virus, with antisera to the egg-grown vaccine strains A/Wisconsin/67/2005 and A/Hiroshima/52/2005 (Tables 4 and 5). As among viruses isolated earlier in 2006, many of the viruses isolated during this period gave low HI titres with all the ferret antisera included in the tests. Antisera to alternative recent reference viruses, such as A/Hiroshima/33/2006 and A/Sendai/F131/2006 (provided by NIID, Tokyo) (Table 4) or A/Nepal/921/2006 (provided by CDC), did not give significantly different results to those obtained with some of the other reference antisera. No consistent correlation was observed between differences in patterns of HI reactivity and particular amino acid changes in HA.

With few exceptions, HA sequences fell within Clade 1, represented by A/Wisconsin/67/2005 and A/Hiroshima/52/2005 and characterized by the common amino acid changes S193F and D225N. Additional changes distinguished 4 principal subgroups among recent isolates (Figure 3, Table 6). In general sequences of the NAs of these viruses fell into corresponding subgroups (Figure 4), one of which corresponded to sequences of the NAs of viruses with Clade 2 HAs, isolated during early 2006 and represented by A/Berlin/2/06. Common amino acid differences from the NAs of A/Wisconsin/67/2005 and A/Hiroshima/52/2005 included L370S together with H150R, V194I, Y310H and S372L, or with N43S and S315R (Table 6). Few HA sequences fell within Clade 2; they were characterized by the changes S45N, D53N, V112I, K173E, S199P and G275D. Their NA sequences were similar to those of Clade 2 viruses isolated earlier in 2006.

M2 sequences of selected viruses showed that, as for viruses isolated earlier in 2006, most viruses with Clade 2 HAs lacked amantadine resistance mutations. In contrast to earlier Clade 1 viruses, the majority of which possessed the amantadine resistance mutation S31N, approximately half of the recent viruses studied lacked known resistance mutations. There was as yet insufficient data to conclude whether the different HA sequence variants vary in their sensitivity to amantadine.

B viruses

Few B viruses isolated during this period were received for characterization. The B/Victoria lineage viruses were antigenically closely related to B/Hong Kong/45/05, the vaccine strain B/Malaysia/2506/04 and the more recent reference strain B/Victoria/304/06 (Table 7). Their HA sequences were close to that of B/Malaysia/2506/04 (Figure 5).

The B/Yamagata lineage viruses were antigenically closely related to the recent reference viruses B/Florida/7/04 and B/Egypt/144/05 (Table 8). HA sequences of recent B/Yamagata lineage isolates were similar to those isolated earlier during 2006; several were characterized by an additional amino acid change G230S (Figure 6). NA sequences of B/Victoria- and B/Yamagata- lineage viruses were similar to those of viruses isolated earlier during 2006 (Figure 7).

Acknowledgments

In addition to data shared among the other 3 Collaborating Centres, sequence data for viruses isolated during this period were received from Influenza Centres in Austria,

Denmark, Finland, France, Germany, Hong Kong, Italy, Norway, Portugal, Sweden and the UK (indicated in Figures by suffixes).

Table 1: Human influenza isolates, September 2006 - present

Country	H1N1		H3N2	B	
	A/NC	A/Sol Is		Yamagata	Victoria
	20/99	3/06		lineage	lineage
September					
Madagascar	10		3		
October					
France			1		
Madagascar	1		10		
Netherlands		1			
United Kingdom		1			
November					
Algeria			2		
Belgium		1	1		
France		1	5		
Israel			1		
Italy			1	1	
Madagascar	1		10		1
Morocco	1		4		
Norway		2	1		
Spain			1		
Sweden			4		
United Kingdom			2		
December					
Belgium			6		
Bulgaria	1	1			
Egypt			1		
France		1	23		
Germany	1	1			
Greece			1	1	
Israel			19		
Italy		3	4		
Madagascar					1
Morocco	2		13		
Netherlands	2		1		
Norway		4	4		
Russia		1			1
Spain	2		3		
Sweden		1	3		
Switzerland			6		
January					
Belgium			1		
Bulgaria			1		
Denmark			4		
Finland			5		
France	3	1	9		
Germany	4	1	7		1
Greece			6		
Italy		15	19		
Latvia			2		
Netherlands	1			1	
Russia	6				
Portugal			22		
Slovakia			10		
Spain	1		9	1	
Sweden			2		
Switzerland			2		
February					
Italy		2	4		
Total = 314	36	37	233	4	4
	73 (23%)		(74%)	8 (3%)	

Table 2. Antigenic analyses of influenza A H1N1 viruses

Viruses	Collection Date		Haemagglutination inhibition titre ¹						
			Post infecton ferret sera						
			A/NC 20/99	A/Thess 24/05	A/Egypt 39/05	A/Fuk 97/06	A/HK 2652/06	A/SI 3/06	A/Fuk 141/06
A/New Caledonia/20/99	9.6.99	Ex	320	640	1280	160	80	160	80
A/Thessaloniki/24/05	3.2.05	E2 \3	640	1280	1280	320	160	320	160
A/Egypt/39/2005	25.11.05	E1 \4	320	640	1280	160	80	160	160
A/Fukushima/97/2006	23.1.06	MDCK4 \1	<	40	160	1280	1280	640	640
A/Hong Kong/2652/2006	17.7.06	MDCK2 \1	40	<	40	320	640	320	320
A/Solomon Islands/3/2006	21.8.06	E3 \2	40	80	160	640	320	320	640
IVR-145 (A/SI/3/2006)		Ex \1	80	160	160	640	320	320	640
A/Fukushima/141/2006	7.2.06	E4 \2	80	80	640	640	640	640	640
A/Madagascar/2649/2006	6.11.06	MDCK1 \1	160	320	640	80	40	nd	nd
A/Morocco/229/2006	18.12.06	MDCKx \1	160	320	640	80	40	80	80
A/Netherlands/2/2007	2.1.07	x, MDCK2 \1	320	1280	1280	160	80	80	80
A/Niedersachsen/1/07	12.1.07	MDCKx \1	320	1280	1280	320	80	160	80
A/St. Petersburg/8/2007	16.1.07	MDCK1 \1	320	640	640	80	80	80	40
A/Barcelona/202/2007	18.1.07	MDCK0 \1	320	1280	1280	160	80	80	40
A/Lyon/291/2007	22.1.07	MDCK2 \1	160	320	1280	80	40	80	80
A/Genoa/8/2007	5.2.07	MDCK2 \1	160	320	1280	160	40	80	160
A/England/555/2006	27.10.06	MDCK2 \3	<	<	80	160	160	nd	nd
A/Paris/160/2007	Nov-06	MDCKx \1	<	<	40	640	640	320	320
A/Belgium/SP002/07	10.11.06	LLCMK1, MDCK2 \2	<	80	40	160	160	160	160
A/Parma/11/2006	Dec 06	MDCK2 \1	<	40	80	640	640	320	320
A/Norway/2285/2006	7.12.06	MDCK1 \1	<	40	160	640	320	nd	nd
A/Stockholm/23/06	12.12.06	MDCK1 \1	<	40	80	640	640	320	320
A/Baden-Wuerttemberg/2/07	Jan-07	MDCKx \1	40	40	160	320	160	320	640
A/Napoli-ISS/4/07	Jan-07	MDCK1 \1	40	160	160	640	320	320	320
A/Trieste/14c/2007	Jan-07	MDCKx \1	40	80	160	320	320	320	320
A/Lyon/30/2007	4.1.07	MDCK2 \1	<	<	80	1280	320	320	320
A/Trieste/34c/2007	Feb-07	MDCKx \1	40	80	80	320	160	320	320

1. < : <40

Figure 1. Phylogenetic comparison of H1 HAs

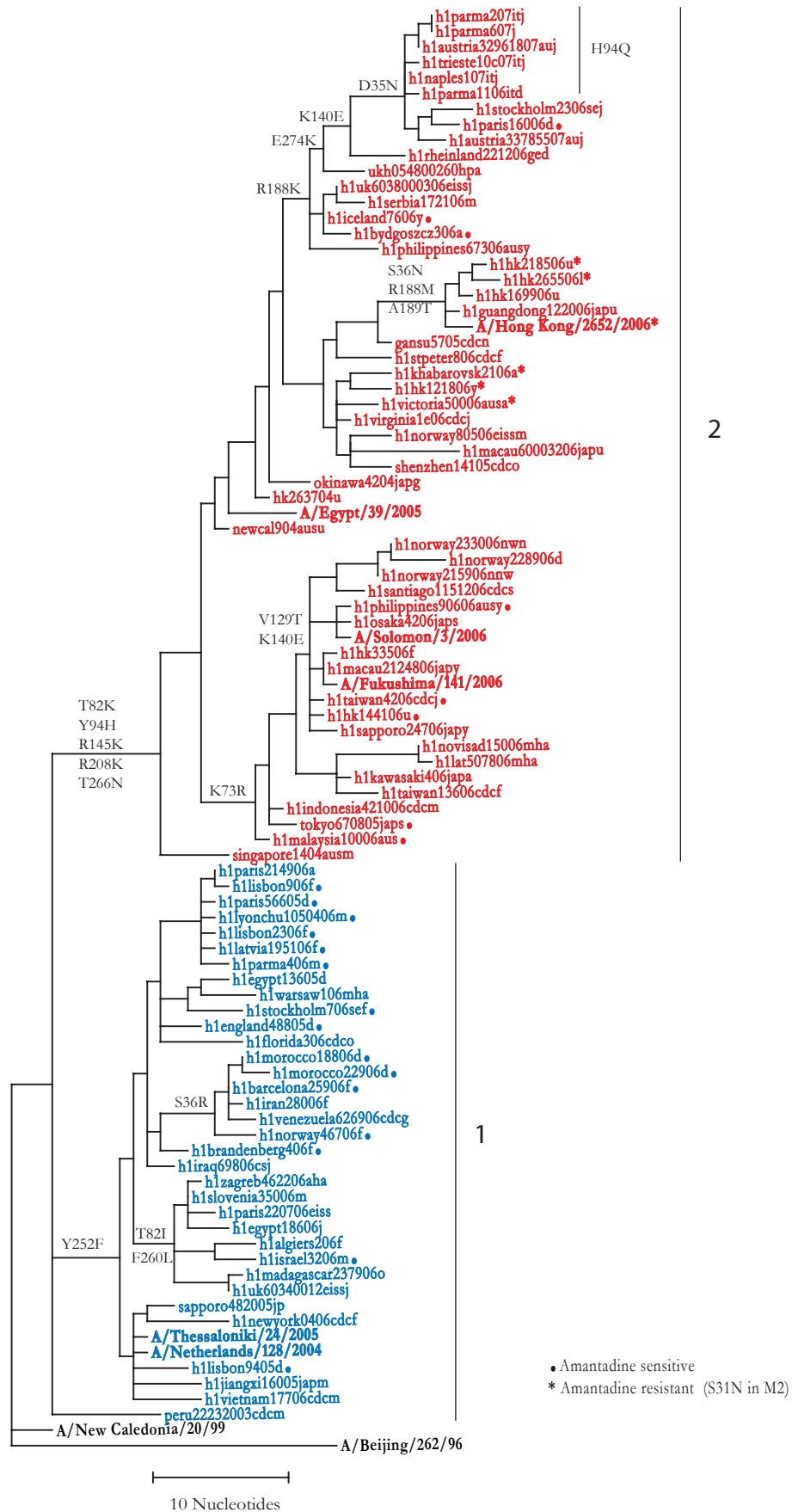


Figure 2. Phylogenetic comparison of N1 NAs

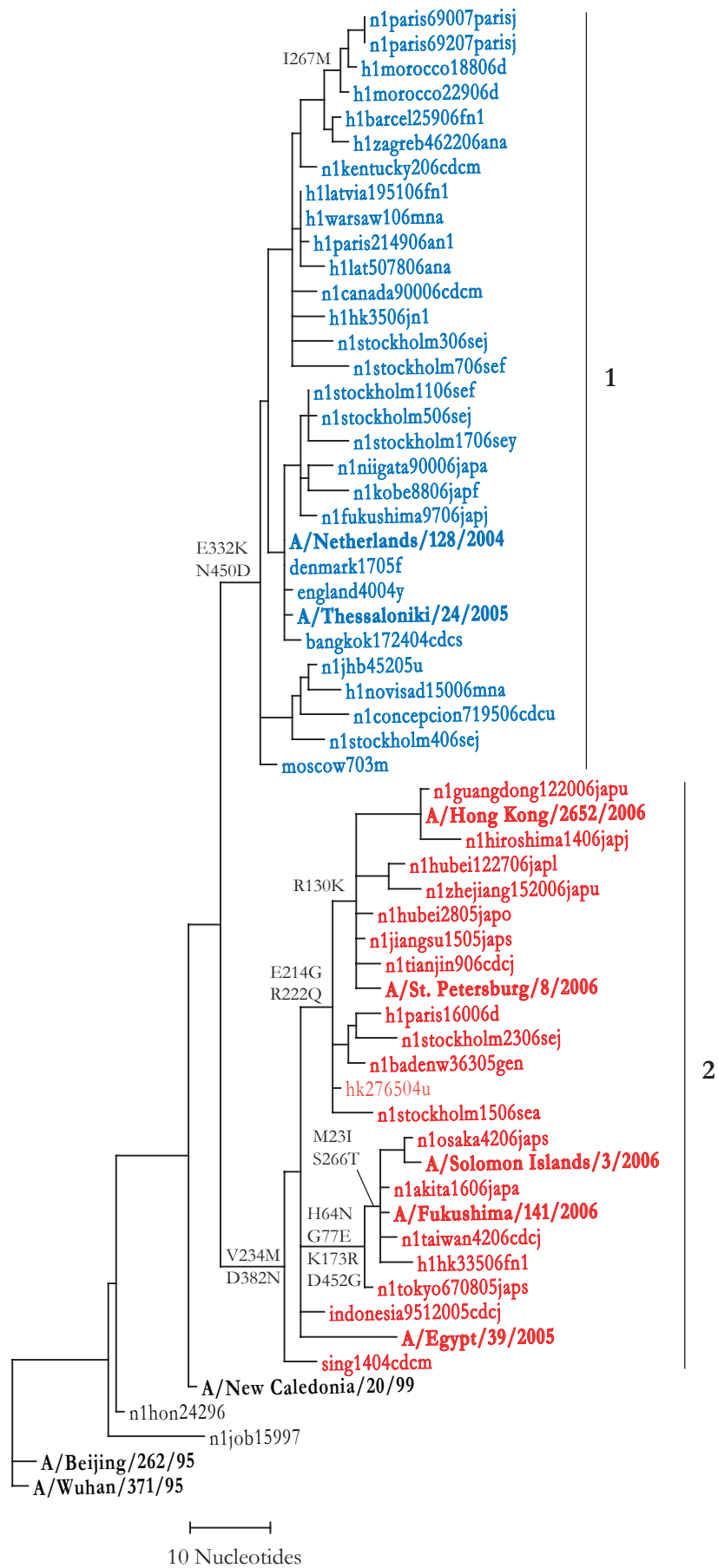


Table 3. Common acid changes characteristic of the principal H1N1 sequence variants

Clade	Representative strain	Amino acid changes ¹	
		HA	NA
1	A/Thessalonika/24/05	Y252F (S36R) (T82I) (F260L)	E332K N450D (I267M)
2	Common changes	T82K Y94H R145K R208K T266N	V234M D382N
	A/Solomon Islands/3/06	K73R V129T K140E (P270S) (D272G)	M23I H64N G77E K173R S266T D452G
	A/Hong Kong/2652/06	S36N R188M A189T	R130K E214G R222Q
	A/Paris/160/06	D35N K140E R188K E274K (H94Q)	E214G R222Q G354D

1. Relative to A/New Caledonia/20/99, including common differences: HA, V165A and W251R; NA, V48I.

Table 4. Antigenic analyses of influenza A H3N2 viruses

Viruses	Isolation Date		Haemagglutination inhibition titre ¹								
			Post infection ferret sera								
			A/Cal	A/HK	A/Hiro	A/Wis	A/Slov	A/Berlin	A/Lyon	A/Hiro	A/Send
			7/04	4443/05	52/05	67/05	134/06	2/06	636/06	33/06	F131/06
			F31/05	F24/05	F2/06	F6/06	F9/06	F8/06	F12/06	JAP	JAP
A/California/7/2004	16.9.04	SpfCk3, E3 \3	5120	640	640	1280	320	640	1280	320	640
A/Hong Kong/4443/2005	24.6.06	MDCK2 \6	1280	1280	1280	2560	640	80	320	640	640
A/Hiroshima/52/2005	25.11.06	E3 \3	640	1280	2560	5120	160	80	320	1280	640
A/Wisconsin/67/2005	31.8.05	SpfCk3E3 \2	1280	1280	2560	5120	640	80	640	320	320
A/Slovakia/134/2006	6.2.06	MDCK1 \3	1280	2560	640	5120	320	80	1280	160	160
A/Berlin/2/2006	6.2.06	MDCKx \3	320	320	320	320	80	80	80	640	320
A/Lyon/636/2006	15.2.06	MDCK2 \4	1280	320	1280	320	80	320	640	640	640
A/Hiroshima/33/2006	23.1.06	E5 \2	320	320	320	320	160	<	160	640	320
A/Sendai/F131/2006	1.8.06	E3 \2	320	1280	640	640	160	40	320	640	640
A/Algeria/G138/2006	5.11.06	MDCK1 \1	320	320	320	320	40	40	80	—	—
A/Madagascar/2670/2006	10.11.06	MDCK1 \1	640	640	640	2560	160	80	640	—	—
A/Greece/624/2006	14.12.06	MDCK1 \1	160	160	320	160	40	40	<	80	80
A/Madagascar/2700/2006	14.11.06	MDCK1 \1	640	320	320	1280	80	80	160	—	—
A/Madagascar/2699/2006	14.11.06	MDCK1 \1	640	320	320	640	80	80	160	—	—
A/Umea/7/06	15.11.06	MDCK1 \1	80	160	80	160	80	20	10	80	<
A/Madagascar/2761/2006	17.11.06	MDCK2 \1	640	640	320	2560	160	80	640	—	—
A/Algeria/G273/2006	22.11.06	MDCK1 \1	80	80	160	40	40	<	<	—	—
A/Marseille/1215/06	22.11.06	Px+P1MDCK \1	640	640	320	160	160	20	160	320	160
A/Bordeaux/1276/06	30.11.06	P3MDCK \1	320	640	640	320	320	40	320	640	320
A/Paris/395/2007	Dec-06	MDCKx \1	320	320	320	320	160	80	320	320	160
A/Parma/9/2006	Dec-06	MDCK1 \1	160	640	640	160	80	20	40	160	160
A/Norway/2250/2006	1.12.06	MDCK1 \1	320	320	320	1280	80	40	160	—	—
A/Norway/2253/2006	4.12.06	MDCK1 \1	320	320	160	640	80	40	80	—	—
A/Norway/2252/2006	4.12.06	MDCK1 \1	160	160	160	320	40	40	80	—	—
A/Switzerland/6936/2006	4.12.06	MDCKx \2	1280	640	320	160	80	80	160	320	160
A/Egypt/99/2006	6.12.06	E2 \1	640	2560	1280	1280	320	40	640	640	640
A/Belgium/SP005/07	19.12.06	LLCMK1, MDCK2 \2	320	640	320	640	320	80	320	640	320
A/Belgium/SP007/07	25.12.06	LLCMK1, MDCK2 \2	80	80	160	80	40	10	<	40	<
A/Lyon/1359/06	27.12.06	P2MDCK \1	160	320	160	80	40	10	40	160	80
A/Belgium/SP003/07	31.12.06	LLCMK1, MDCK2 \1	640	640	320	1280	160	20	640	640	320
A/Lyon/1324/06	20.12.06	P2MDCK \1	320	320	320	320	160	20	80	320	160
A/Lyon/1312/06	20.12.06	P2MDCK \1	320	640	320	320	160	20	80	320	160
A/Belgium/SP009/07	7.1.06	LLCMK1, MDCK2 \2	160	320	320	160	160	40	160	160	80

1. <, <40

Table 5. Antigenic analyses of influenza A H3N2 viruses

Viruses	Isolation Date		Haemagglutination inhibition titre ¹						
			Post infecton ferret sera						
			A/Cal	A/HK	A/Hiro	A/Wis	A/Slov	A/Berlin	A/Lyon
			7/04	4443/05	52/05	67/05	134/06	2/06	636/06
			F31/05	F24/05	F2/06	F6/06	F9/06	F8/06	F12/06
A/California/7/2004	16.9.04	SpfCk3, E3 \3	5120	640	320	320	80	640	1280
A/Hong Kong/4443/2005	24.6.06	MDCK2 \6	640	640	320	320	80	80	160
A/Hiroshima/52/2005	25.11.06	E3 \3	640	1280	5120	2560	320	80	640
A/Wisconsin/67/2005	31.8.05	SpfCk3E3 \2	1280	2560	2560	5120	320	80	640
A/Slovakia/134/2006	6.2.06	MDCK1 \3	1280	1280	1280	2560	320	80	1280
A/Berlin/2/2006	6.2.06	MDCKx \3	640	320	320	160	80	160	320
A/Lyon/636/2006	15.2.06	MDCK2 \4	5120	1280	1280	1280	160	640	640
A/Nepal/921/2006									
A/Norway/2253/2006	4.12.06	MDCK1 \1	320	160	640	320	320	80	80
A/Switzerland/6936/2006	4.12.06	MDCKx \2	1280	640	320	160	80	80	160
A/Egypt/99/2006	6.12.06	E2 \1	640	2560	1280	1280	320	40	640
A/Belgium/SP009/07	7.12.06	LLCMK1, MDCK2 \2	160	320	320	160	160	40	160
A/Israel/73/2006	24.12.06	MDCK2 \2	320	320	320	160	80	40	<
A/Netherlands/363/2006	24.12.06	MDCK1 \1	160	640	320	320	160	20	320
A/Barcelona/134/2006	28.12.06	MDCK1 \1	160	320	320	320	80	80	80
A/Parma/4/07	Jan-07	MDCK1 \1	80	160	160	320	40	20	20
A/Sofia/319/07	Jan-07	E2 \1	1280	640	320	640	160	40	640
A/Stockholm/25/06	2.1.07	MDCK0 \1	160	160	160	80	40	20	<
A/Switzerland/7567/2007	9.1.07	MDCKx \2	320	320	320	320	80	20	320
A/Latvia/345/2007	9.1.07	MDCK1 \2	160	160	160	80	20	10	<
A/Bratislava/46/07	9.1.07	E1 \1	1280	640	320	640	160	20	640
A/Denmark/1/2007	12.1.07	MDCK2 \1	320	640	1280	640	320	40	320
A/Berlin/4/07	15.1.07	MDCKx \1	320	640	640	640	160	80	320
A/Finland/800/07	16.1.07	MDCK2 \3	160	320	320	320	80	20	20
A/Thessaloniki/26/07	17.1.07	MDCK2 \1	160	320	640	320	40	20	40
A/Lisbon/14/2007	23.1.07	MDCK2 \1	1280	640	640	640	160	40	640
A/Lyon/323/2007	22.1.07	MDCK2 \1	160	320	320	320	80	40	40
A/Serres/77/07	25.1.07	E1 \1	80	160	640	640	80	40	160
A/Valladolid/8/2007	25.1.07	MDCK1 \2	1280	640	640	1280	160	40	640
A/Trieste/29c/2007	Feb-07	MDCKx \1	320	640	640	640	320	80	160

1. <, <40

Figure 3. Phylogenetic comparison of H3 HAs

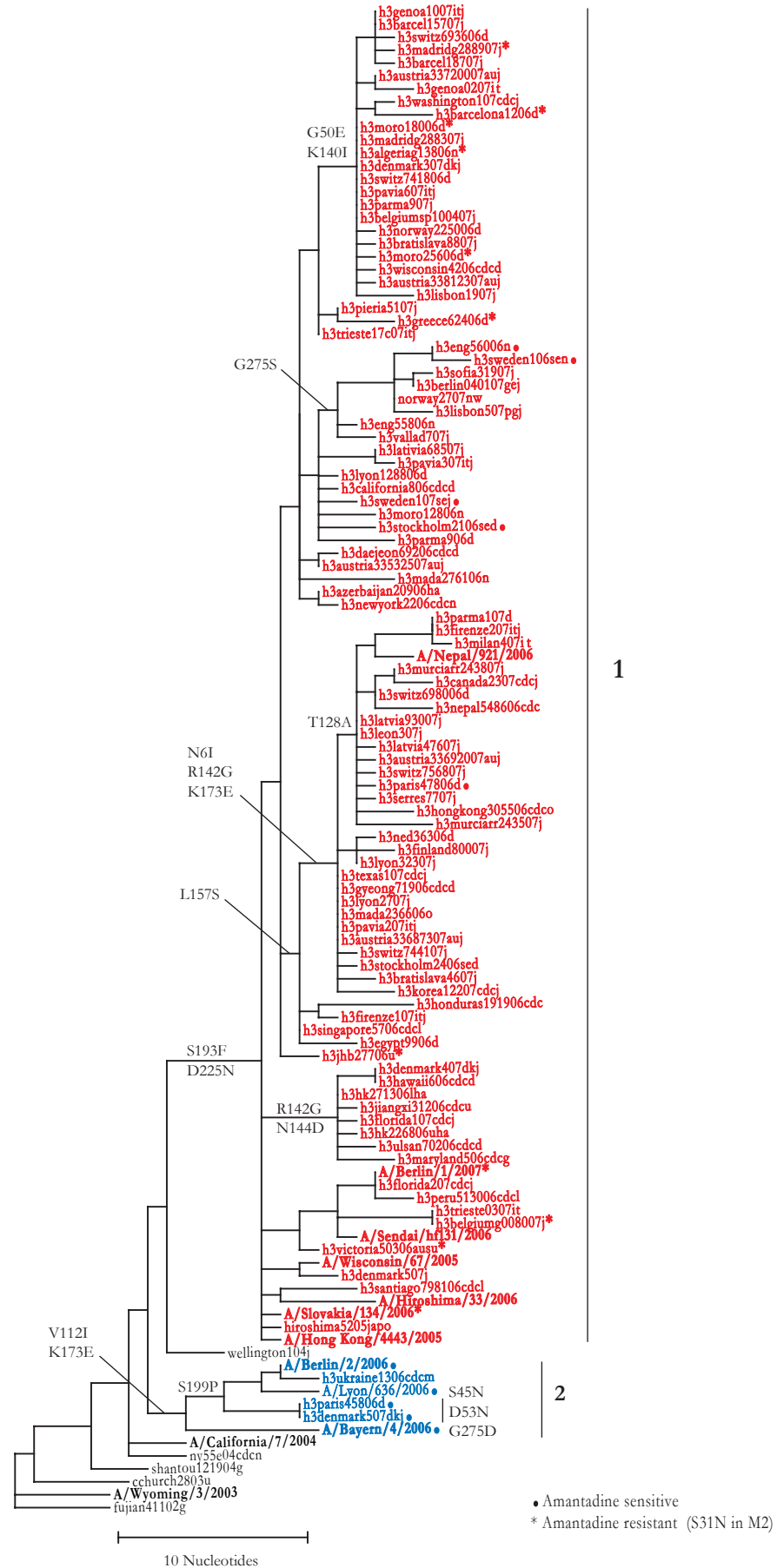


Figure 4. Phylogenetic comparison of N2s

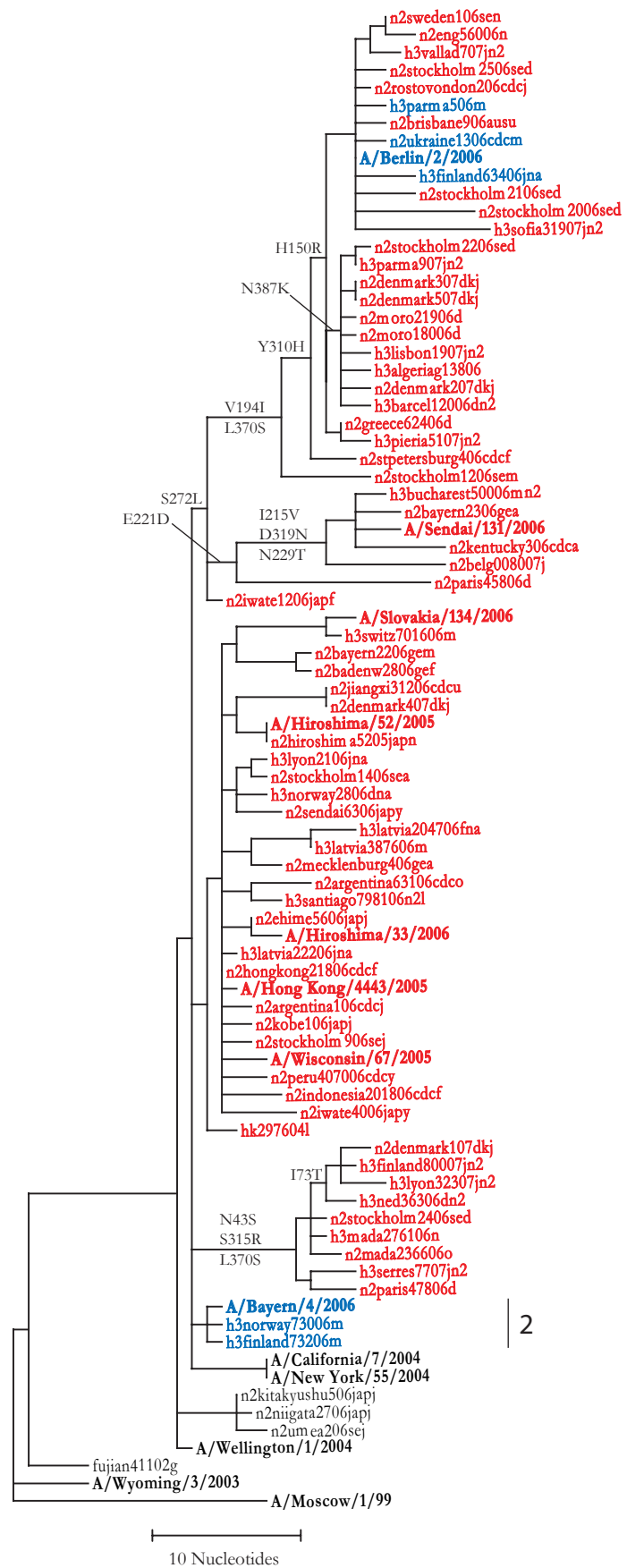


Table 6. Common amino acid changes characteristic of the principal H3N2 sequence variants

Clade	Representative strain	Amino Acid changes ¹	
		HA	NA
1-1	A/Hong Kong/2713/06	S193F D225N R142G N144D	D93N N86K H150R K296R I307M S335G
1-2	A/Nepal/921/06	S193F D225N N6I R142G L157S R173E (T128A)	N43S S315R L370S (I73T) (T67I)
1-3	A/Morocco/180/06	S193F D225N G50E K140I	H150R V194I Y310H L370S S372L N387K
1-4	A/England/560/06	S193F D225N G275S	H150R V194I Y310H L370S S372L
2	A/Berlin/2/06	V112I K173E S199P (S45N) (D53N) (G275D)	H150R V194I Y310H L370S S372L

1. Relative to A/California/7/04.

Table 7. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date		Haemagglutination inhibition titre ¹					
			B/Shan ² 7/97	Post infection ferret sera				
				B/Shan 7/97	B/Bris 32/02	B/HK 45/05	B/Mal 2506/04	B/Vic 304/06
B/Shandong/7/97		Ex	640	320	320	160	320	320
B/Brisbane/32/2002	2.7.02	E2 \6	320	160	160	160	160	160
B/HK/45/2005	7.2.05	MDCK2 \4	640	160	160	160	320	320
B/Malaysia/2506/2004	6.12.04	E3 \3	320	160	160	160	320	320
B/Victoria/304/2006	6.6.06	E2 \2	320	160	320	320	320	320
B/Hong Kong/840/2006	17.6.06	MDCK2 \2	320	40	<	80	80	nt
B/Hong Kong/859/2006	17.7.06	MDCK2 \3	320	40	<	80	80	nt
B/Bangladesh/5278/2006	19.7.06	E2 \1	160	80	160	80	80	80
B/Texas/39/2006	5.11.06	E3 \1	160	80	160	160	160	160
B/Madagascar/2828/2006	29.11.06	MDCK1 \1	640	160	80	320	80	80
B/Madagascar/2866/2006	5.12.06	MDCK1 \1	640	160	160	80	320	320
B/Astrakhan/1/2006	28.12.06	MDCK1 \1	640	160	40	160	160	160
B/Baden-Wuerttemberg/1/07	16.01.07	MDCKx \1	640	160	40	160	160	160

¹ <, <10

² hyperimmune sheep serum

Table 8. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Collection date		Haemagglutination inhibition titre*				
			Post infection ferret sera				
			B/Sich 379/99	B/Shai 361/02	B/Jiang 10/03	B/Eg 144/05	B/Fl 7/04
B/Sichuan/379/99		Ex	160	320	20	160	320
B/Shanghai/361/2002	12.6.02	E3 \4	320	640	80	320	640
B/Jiangsu/10/2003	12.9.03	E4 \3	40	40	320	80	80
B/Egypt/144/2005	5.1.05	E2 \2	320	640	40	320	640
B/Florida/7/2004	1.10.05	E3 \2	160	320	40	320	640
B/Hong Kong/849/2006	30.6.06	MDCK2 \1	160	160	80	160	320
B/Hong Kong/864/2006	24.7.06	MDCK2 \3	160	80	40	160	320
B/Taiwan/79/2006	27.9.06	C3 \1	80	320	40	320	320
B/Roma-Iss/1/2007	Nov 06	MDCK 1 \1	80	160	160	160	160
B/Greece/628/2006	15.12.06	MDCK2 \1	80	160	160	160	160
B/Netherlands/1/2007	4.1.07	MDCK1 \1	160	160	80	320	320
B/Valladolid/1/2007	16.1.07	MDCK1 \2	160	160	80	320	640

¹ <, <40

Figure 5. Phylogenetic comparison of B HAs (Victoria lineage)

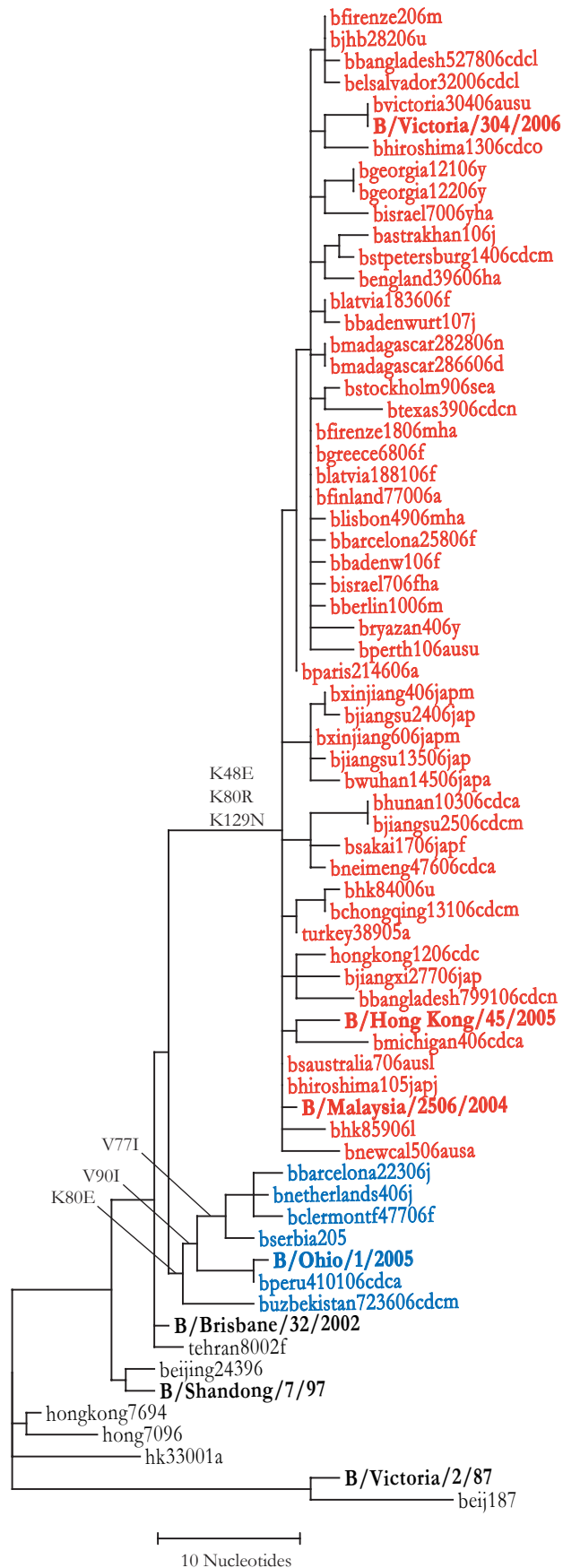


Figure 6. Phylogenetic comparison of B HAs (Yamagata lineage)

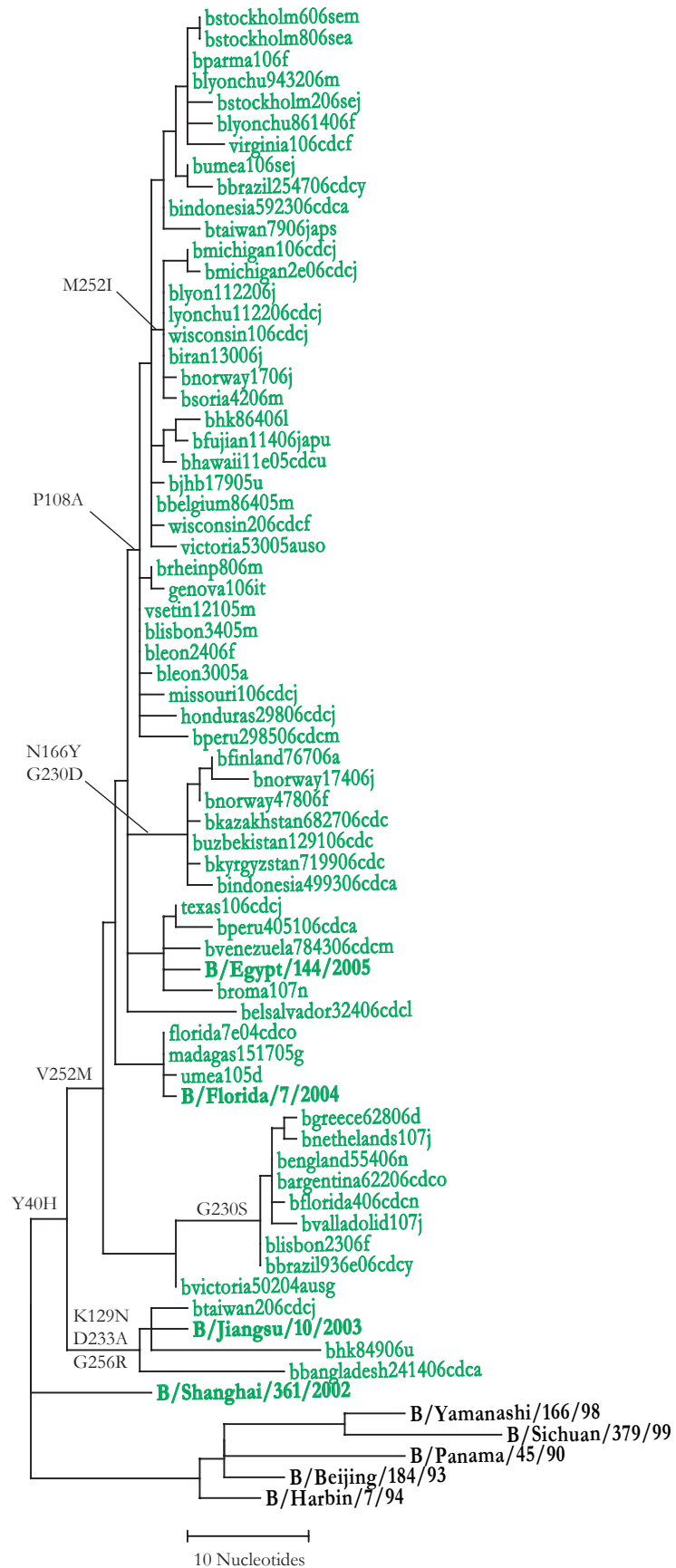


Figure 7. Phylogenetic comparison of B NAs

